

Homework (Habanero)

- Q1.** What is the slope of a line parallel to $y = 2x + 1$?
(A) $m = 1$
(B) $m = 2$
(C) $m = 3$
(D) $m = 4$
(E) $m = 5$
- Q2.** A patient's temperature is modeled by $T(t) = 2t + 98.6$. What is the equation of a line parallel to this model?
(A) $T(t) = 2t + 100$
(B) $T(t) = 2t - 100$
(C) $T(t) = t + 98.6$
(D) $T(t) = 3t + 98.6$
(E) $T(t) = -2t + 98.6$
- Q3.** A biologist is studying the growth of cells. The rate of growth is modeled by $y = 3x + 2$. What is the slope of a line parallel to this model?
(A) $m = 1$
(B) $m = 2$
(C) $m = 3$
(D) $m = 4$
(E) $m = 5$
- Q4.** What is the equation of a line parallel to $y = -2x + 1$ that passes through $(2, 3)$?
(A) $y = -2x + 7$
(B) $y = -2x - 1$
(C) $y = 2x + 1$
(D) $y = -2x + 5$
(E) $y = 2x - 1$
- Q5.** A doctor prescribes a dosage of $D(x) = -0.5x + 100$ milligrams. What is the equation of a line parallel to this model?
(A) $D(x) = -0.5x + 90$
(B) $D(x) = -0.5x - 100$
(C) $D(x) = 0.5x + 100$
(D) $D(x) = 0.5x - 100$
(E) $D(x) = -1x + 100$
- Q6.** Which of the following lines is parallel to $y = 1.5x - 2$?
(A) $y = 1.5x + 1$
(B) $y = 2x - 1$
(C) $y = -1.5x + 2$
(D) $y = x - 2$
(E) $y = x + 2$
- Q7.** A line has a slope of $m = 2$ and passes through $(1, 3)$. What is the equation of a line parallel to this line?
(A) $y = 2x + 1$
(B) $y = 2x - 1$
(C) $y = 2x + 3$
(D) $y = x + 2$
(E) $y = -2x + 4$
- Q8.** The cost of a medical treatment is modeled by $C(x) = 0.8x + 50$. What is the equation of a line parallel to this model?
(A) $C(x) = 0.8x - 50$
(B) $C(x) = 0.8x + 60$
(C) $C(x) = -0.8x + 50$
(D) $C(x) = x + 50$
(E) $C(x) = -x - 50$

Open Ended Questions (FRQ)

- Q9.** A line has a slope of $m = -1.2$ and passes through $(2, 4)$. Find the equation of a line parallel to this line that passes through $(1, 2)$.
- Q10.** The growth rate of a cell culture is modeled by $y = 2.5x + 1$. Find the equation of a line parallel to this model that passes through $(3, 9)$.

Chef's Tips

- Emphasize the importance of identifying the slope of parallel lines.
- Encourage students to use real-world examples, such as medicine dosage or cell growth.
- Remind students to check their work and provide clear explanations for free response questions.